

AI Act Risk Classifier Pro

Detailed Project Documentation: Idea, Business Problem, AI-Assisted Development Process, Technical Architecture, MVP, Real-Life Roadmap, and Test Cases

Item	Description
Product	AI Act Risk Classifier Pro
Purpose	Classify AI systems into EU AI Act risk tiers and generate compliance guidance
Frontend libraries	React, TypeScript, Tailwind CSS, Vite, lucide-react
AI tools used	Codex for product generation/refactoring; ChatGPT for prompt enhancement and EU AI Act legal research support
QA status	PASS - 5 passed, 0 failed
Generated	27 April 2026, 18:00

Academic disclaimer: this product is a proof of concept and educational decision-support tool. It does not provide legal advice and should not replace qualified legal review.

1. Why This Idea Was Chosen

The idea was chosen because the EU AI Act creates a practical problem: many teams can explain what an AI product does, but they cannot easily translate that product description into a risk tier, obligations, evidence requirements, and a report. This gap is especially visible for startups, students, small companies, and product teams that do not have immediate access to legal specialists.

- AI regulation is growing, but many AI builders are technical or business-focused rather than legal experts.
- The AI Act uses a risk-based model, which means the answer depends on context, affected people, data, deployment domain, and system behavior.
- A simple chatbot-style answer is not enough because compliance needs auditability, evidence, citations, and repeatable reasoning.
- A guided compliance product is a strong LegalTech concept because it turns legal complexity into a workflow that can be demonstrated, improved, and sold.

2. Problem The Product Solves

The core problem is not only classification. The real problem is producing a structured and defensible compliance story. A user needs to know: What is the risk tier? Why was it classified that way? What laws or annexes triggered the result? What evidence is missing? What should be changed to reduce risk? What should be included in a report?

Observed Problem	Product Response
Users do not know legal terminology	The UI asks plain-language product questions instead of legal questions.
Risk depends on context	The engine maps purpose, sector, impact, safeguards, data, and evidence to legal categories.
Compliance must be explainable	The result includes a visible reasoning pipeline and hidden expert citations.
Legal output needs evidence	The app accepts model cards, datasets, logs, and technical documentation.
Users need next steps	The app produces checklists, high-risk workflow tasks, and risk-reduction suggestions.

3. Tools, Libraries, and AI Assistance Used

The project combines normal frontend engineering tools with AI-assisted development and legal research support. The role of each tool is separated clearly below.

Tool or Library	How It Was Used	Why It Was Chosen
React	Built the component-based web interface: homepage, wizard, dashboard, and report viewer.	Fast and flexible for building interactive single-page apps and reusable UI components.
TypeScript	Typed inputs, rules, risk tiers, evidence documents, citations, and report outputs.	Reduces bugs and enforces consistency in a complex compliance product.
Tailwind CSS	Created a responsive, calm, modern UI with utility classes.	Supports quick design iteration while keeping styling consistent.
Vite	Development server and production build tool.	Fast local development and simple deployment to free static hosts.
lucide-react	Provided icons for actions, status, evidence, testing, reports, and navigation.	Clean icon system that makes the UI easier to understand.
ReportLab	Generated the professor-facing PDF documentation.	Reliable Python library for structured PDF output.
Codex	Generated and refactored the product code, legal engine models, and QA harnesses.	Boosts QA aims, high-level product scripts, and working React/TypeScript code.
ChatGPT	Enhanced the initial prompt, helped structure the product requirements, and supported legal research.	Provides legal AI for the AI Act ideas, source code, and research.

Important credibility note: ChatGPT was used as a research and summarization assistant, but the legal logic was aligned with official EU AI Act sources, European Commission public guidance, and class materials. The app intentionally includes disclaimers and uncertainty warnings.

4. Step-by-Step Project Process

Step	What Was Done	Output
1. Idea framing	Identified that AI Act classification is hard for non-lawyers and useful for startups/classroom use.	Legal Tech upsides/consensus.
2. Prompt enhancement	Used ChatGPT to make the initial product request more complete and structured.	Clear requirements for UI, legal engine, evidence, QA, and report.
3. Legal source collection	Used ChatGPT to collect and summarize EU AI Act concepts, then asked for official EU Commission Annexes categories, and examples.	Relevant official EU Commission Annexes categories, and examples.
4. Product design	Converted legal requirements into a beginner-friendly flow: Describe, Simplify, Assess, Review, Export.	Simplify, Assess, Review, Export legal-form complexity.
5. Code generation	Used Codex to build React/TypeScript components, legal modules, writing templates, report view, and styling.	Writing templates, report view, and styling.
6. Evidence engine	Added local text evidence scanning for sensitive attributes, oversight, data misuse, model bias signals.	Evidence retrieval, model bias signals.
7. QA and fixing	Built a full-cycle QA harness and fixed bugs found during testing.	5 passed, 0 failed full-cycle QA result.
8. Documentation	Generated professor documentation explaining business, tech, legal, and R&D MVP roadmap, and tests.	High R&D MVP roadmap, and tests.

5. MVP Definition

The MVP is the smallest credible version that demonstrates the full compliance workflow from intake to report. The current product already covers the MVP scope.

- Homepage that clearly explains the product value.
- Guided wizard using simple product questions.
- Rule-based EU AI Act risk classification engine.
- Scenario playground for classroom demonstration.
- Evidence upload and risk signal extraction.
- Optional model-behavior test hooks.
- Confidence and uncertainty bars.
- Legal citations hidden behind expert-mode controls.
- High-risk compliance checklist and conformity workflow.
- Audit trail with versioned assessments.
- Printable report and professor documentation.

6. How The Product Works

From the user's perspective, the product is deliberately simple. The user does not start by reading legal text. Instead, they describe what the AI does, identify obvious risk signals, add evidence, review uncertainty, and export the result.

User Step	Plain-Language Purpose	Engine Action
Describe	Collect product facts: purpose, sector, users, data, EU	Builds a factual profile for legal mapping.
Identify	Select obvious risks using ordinary-world wording.	Maps answers to Article 5, Annex III, and Article 50 logic.
Assess	Add impact, safeguards, evidence files, and optional checks.	Extracts evidence signals and runs model-check hooks.
Review	Shows summary, missing info, and uncertainty before decision.	Presents final input object.
Export	Runs classification and creates report/evidence bundle.	Generates result, checklist, citations, audit record, and report page.

7. Legal and Compliance Logic

- Article 5 prohibited-practice screen: harmful manipulation, exploitation of vulnerabilities, social scoring, certain biometric/emotion uses, and criminal-risk profiling patterns.
- Annex III high-risk screen: employment, education, law enforcement, migration/border, healthcare, critical infrastructure, essential services, biometrics, justice, and democratic processes.
- Transparency screen: chatbot interaction, synthetic media, deepfakes, generated content, emotion recognition, and biometric categorisation disclosures.
- Annex I-V support: clause summaries for product-safety context, serious offence context, high-risk categories, technical documentation, and EU declaration-of-conformity elements.
- Uncertainty model: shows evidence support, real-world stability, information completeness, and contradiction level as visual bars.

8. Real-Life Implementation Plan

Phase	Goal	What Would Be Added
POC / Class Demo	Show the complete idea and prove the work	Static deployment, scenario demos, PDF report, QA evidence.
Pilot	Test with students, founders, or small legal/consulting teams	User feedback, improved legal wording, better evidence templates, export branding
Professional MVP	Make it usable for real teams.	Accounts, secure workspaces, persistent database, role-based access, report history
Production Compliance	Support serious business use.	Verified legal knowledge base, versioned legal updates, expert review workflow, feedback
Enterprise / Consulting Version	Support lawyers and consultants.	Client workspaces, matter management, review comments, sign-off, permissions, integrations

In real life, this product could help startups screen AI features before launch, help compliance teams prepare internal assessments, help law students understand AI Act risk categories, and help consultants create structured first-pass reports for clients.

9. Test Cases Used

The product includes a test harness that checks the key scenarios and technical flows. These cases show that the system can classify different AI systems, process evidence, generate reports, and preserve audit records.

Case	Input Summary	Expected Result	Why It Matters
AI CV Screening	Ranks job applicants and recommends interview shortlists.	High Risk	Tests employment and recruitment high-risk mapping.
Predictive Policing	Predicts criminal risk from profiling/demographic signals.	Highly Significant Risk	Tests Article 5 stop-sign logic.
Medical Diagnosis AI	Supports clinicians with triage and diagnostic suggestions.	High Risk	Tests healthcare and safety-critical mapping.
Customer Chatbot	Answers support questions and opens tickets with decision rights.	Low Risk	Tests transparency duties and negation handling.
Spam Filter	Classifies emails as spam or safe with user feedback.	Minimal Risk	Tests minimal-risk fallback and baseline citations.
Emotion Detection in Schools	Monitors student attention/emotion from webcam feeds.	Unacceptable Risk	Tests education emotion-recognition prohibition.
Evidence Stress Case	Uploaded text includes sensitive attributes, high-risk information, and legal advice.	High Risk with human input	Tests evidence ingestion, results, model hooks, and uncertainty, and citations.
Custom Minimal Intake	Internal email draft helper with human review and material decision rights.	Minimal Risk	Tests default start user journey.

10. Automated QA Result

Test	Status	Detail
Scenario library produces expected risk tiers	PASS	Passed
Evidence ingestion affects findings, uncertainty, and testing hooks	PASS	Passed
Report manifest includes business, evidence, compliance, citations, and tests	PASS	Passed
Audit layer versions repeated assessments and records changes	PASS	Passed
Minimal custom intake can complete the full classifier path	PASS	Passed

The final verification also passed TypeScript compilation, production build, and local HTTP availability at <http://127.0.0.1:5173/>.

11. Risks, Limitations, and Ethics

- The product is not legal advice and should not be used as the final legal determination for a real deployment.
- Legal rules are summarized for screening and teaching; legal counsel should verify edge cases, exceptions, and sector-specific laws.
- Evidence extraction is keyword-based and should be replaced with stronger document parsing for production.
- Model testing hooks are evidence-based summaries, not full technical audits.
- A production system would need authentication, encrypted storage, role-based access, retention controls, and a verified legal update process.

12. Sources Used for Legal Framing

- Regulation (EU) 2024/1689, official EU AI Act text on EUR-Lex.
- European Commission AI Act overview: risk-based approach, prohibited practices, high-risk examples, transparency duties, GPAI timing, and implementation timeline.
- AI Act Service Desk and class materials for annex structure, high-risk systems, prohibited practices, GPAI, and data-protection context.